# Edge Quantum Guard (EQG): Technical Report on Validation and Cross-Hardware Reproducibility Plan

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EQG is a four-role control stack that improved QAOA solution quality in controlled runs on three IBM processors (ibm_fez, ibm_marrakesh and ibm_kingston); this report states exactly what was measured, what was not, and how the effect will be reproduced and validated on other quantum hardware types.

## 1. System Under Test

EQG operates as an autonomous low-level control layer between the user's workload and the quantum backend.

| Role | Module | Function |
|---|---|---|
| Eyes | AOM / health | Measures friction, regime, and per-qubit health from calibration data and live measurement |
| Reflexes | MORPH | Selects the best connected physical path from qubit health and two-qubit gate error |
| Judgment | Minimal plan + causal verdict | Decides between conditional dynamical decoupling (DD), remap, or hold; issues a causal verdict on each intervention |
| Hands | DD + execution | Inserts CPMG/XY4 sequences in idle windows and executes the workload |

A companion probe, Centinela, runs about 7 seconds of QPU time to classify candidate qubits as stable or degraded, estimate DD benefit, and compute a virtual-Z phase correction.

## 2. Validations Performed on Real Hardware

All results below come from ibm_fez, a real IBM Quantum processor. The workload was QAOA MaxCut on a path graph (N = 8, p = 2). The pre-registered criterion was $z \geq 2$ on $\Delta AR = AR(\text{EQG on}) - AR(\text{EQG off})$.

## 2.1 Controlled A/B runs

| Metric | R1 (EQG first, 7 s/arm) | R2 (baseline first, 15 s/arm) |
|---|---|---|
| Calibration date | 2026-10-10 | 2026-10-12 |
| QAOA shots per arm (approx.) | 2,800 | 25,200 |
| AR, EQG on | 0.8147 ± 0.0027 | 0.8092 ± 0.0009 |
| AR, EQG off | 0.7923 ± 0.0029 | 0.7713 ± 0.0010 |
| ΔAR | +0.0224 ± 0.0040 | +0.0379 ± 0.0014 |
| z-score | 5.7 | 27.7 |
| Retention, on / off | 0.95 / 0.93 | 0.95 / 0.90 |
| P(optimal), on / off | 0.221 / 0.185 | 0.212 / 0.156 |

Both runs exceeded the threshold, and inverting the arm order preserved the sign of the effect. This reduces, but does not remove, the confound of chip drift between sequential arms.

## 2.2 Stability during the run

In R2, AR on the EQG arm drifted +0.0096 (rising) while the baseline drifted −0.0050 (falling). Mean instance stability s was 0.744 (on) versus 0.645 (off) in R2, and 0.716 versus 0.656 in R1.

## 2.3 Attribution from simultaneous snapshots

| Effect on stability s | R1 | R2 |
|---|---|---|
| Layout (MORPH vs naive) | +0.033 | +0.104 |
| DD on vs off | +0.061 | +0.073 |

Both components contributed measurably and significantly; layout typically dominated the solution-quality gain, with DD reinforcing effective coherence.

## 2.4 Reactive behavior (R2)

In the first EQG job the regime was stable and Judgment held DD off. Between jobs a coherence/health signal triggered conditional DD (28 pulses in job 1), and AR on the EQG arm rose from 0.805 to 0.815. The causal verdict was exito=True. This shows a reactive

mechanism, not only a static layout choice at time zero.

## 2.5 Acceptance criteria and job records

Criteria C1–C7 (full stack, budgets, AR and stability series, PRE/MID/POST snapshots, statistical verdict) were met in both runs. R2 job IDs on ibm_fez: PRE db6557o4qg6s73c489eg; A db655aamb58s738b8vdg and db655dkvf2bc73d0m370; MID db655hklf4us73c4qfpg; B db655k4lf4us73c4qft0 and db655nclf4us73c4qg00; POST db655rclf4us73c4qg4g. Outputs per run: report.json, summary.txt, qaoa_stability.png.

## 2.6 Supporting hardware characterization (Centinela, qubit q142, 23 Sept 2026)

Four jobs on ibm_fez showed two discrete coherence states: T2* free of 155 ± 5 μs (good) and 40 ± 4 μs (degraded). DD extended coherence by about 1.4x in the good state and 2.1x in the degraded state, removed about 65% of the excess dephasing, and suppressed measured ZZ coupling by about 93%. Relay (crosstalk) population transfer to neighbors stayed below 0.18% (upper 95% bound) in all runs. This motivates EQG's state-dependent DD decision.

## 2.7 Four-arm ablation on ibm_marrakesh (12 Oct 2026)

A second backend, ibm_marrakesh, gave a controlled ablation of the same QAOA workload (N = 8, p = 2). Four arms were interleaved inside each job in palindromic (ABBA) order to limit chip drift: A = naive layout, C = MORPH layout from calibration, D = MORPH layout with DD forced on, E = full EQG (live measurement, conditional DD, reaction between jobs). Calibration stamp 2026-10-12 05:10 UTC; 18.0 s of QPU time reported by IBM across 4 jobs; 6,000 shots per arm (1,500 shots x 2 repetitions x 2 blocks).

| Arm | Configuration | AR | P(optimal) |
|---|---|---|---|
| A | Naive layout, no DD | 0.7189 ± 0.0023 | 0.116 |
| C | MORPH layout by calibration | 0.7925 ± 0.0020 | 0.182 |
| D | MORPH layout + DD forced | 0.7898 ± 0.0020 | 0.179 |
| E | Full EQG | 0.8126 ± 0.0019 | 0.213 |

Ideal (noiseless) AR for this circuit is 0.8559. The pre-set criterion was $z \geqslant 2$ with a positive sign in all 4 replicas.

| Contrast | What it isolates | ΔAR ± SE | z | Positive replicas | Verdict |
|---|---|---|---|---|---|
| C − A | Layout chosen by calibration | +0.0736 ± 0.0030 | 24.6 | 4/4 | Supported |

| Contrast | What it isolates | ΔAR ± SE | z | Positive replicas | Verdict |
|---|---|---|---|---|---|
| D − C | DD in isolation | −0.0027 ± 0.0028 | −1.0 | 1/4 | Inconclusive |
| E − C | Live measurement + reaction | +0.0200 ± 0.0027 | 7.4 | 4/4 | Supported |
| E − D | Conditional policy vs DD always on | +0.0228 ± 0.0027 | 8.3 | 4/4 | Supported |
| E − A | Full stack vs baseline | +0.0937 ± 0.0029 | 31.9 | 4/4 | Supported |

- Most of the gain (0.0736 of 0.0937) comes from the calibration-based layout. On this chip that day the naive layout [0–7] was poor.
- Live measurement and reaction added a further +0.0200 on top of the calibration layout.
- DD forced on by itself did not improve AR (inconclusive). This differs from the stability-metric gain attributed to DD on ibm_fez, so the DD contribution should be tested separately on each device.
- The conditional policy outperformed always-on DD.
- E's trajectory: at PRE, Judgment chose a remap with DD on; between jobs the regime was stable and it chose no intervention. Because E used a different layout than C, E − C reflects the live-measured remap, not DD.

Layouts: A [0–7]; C [9, 8, 7, 6, 5, 4, 3, 2]; E [92, 91, 90, 89, 88, 87, 97, 107]. Job IDs: PRE db671hklf4us73c4t8ug; block 1 db671m84qg6s73c4b3l0; block 2 db671pkvf2bc73d0osh0; POST db671tklf4us73c4t9h0. Outputs: report.json and ablation.png.

Limits stated in the notebook itself: one run, one calibration cycle, four replicas; palindromic ordering reduces but does not remove drift; "supported" means consistent in this run, not a general law.

## 2.8 Four-arm ablation on ibm_kingston (12 Oct 2026)

The same protocol, workload and pre-set criterion ($z \geq 2$ with a positive sign in all 4 replicas) were repeated on ibm_kingston. Calibration stamp 2026-10-12 08:05 UTC; 16.0 s of QPU time reported by IBM across 4 jobs; 6,000 shots per arm. The result is mixed and is reported as such.

| Arm | Configuration | AR | P(optimal) |
|---|---|---|---|
| A | Naive layout, no DD | 0.7700 ± 0.0019 | 0.137 |
| C | MORPH layout by calibration | 0.8216 ± 0.0018 | 0.235 |
| D | MORPH layout + DD forced | 0.8150 ± 0.0019 | 0.224 |
| E | Full EQG | 0.8157 ± 0.0019 | 0.228 |

| Contrast | What it isolates | ΔAR ± SE | z | Positive replicas | Verdict |
|---|---|---|---|---|---|
| C − A | Layout chosen by calibration | +0.0516 ± 0.0027 | 19.3 | 4/4 | Supported |
| D − C | DD in isolation | −0.0066 ± 0.0026 | −2.5 | 0/4 | Opposite sign |
| E − C | Live measurement + reaction | −0.0059 ± 0.0026 | −2.2 | 0/4 | Opposite sign |
| E − D | Conditional policy vs DD always on | +0.0008 ± 0.0027 | 0.3 | 3/4 | Inconclusive |
| E − A | Full stack vs baseline | +0.0458 ± 0.0027 | 16.9 | 4/4 | Supported |

- Full EQG beat the naive layout by +0.0458, and nearly all of it is the calibration-based layout [50, 51, 58, 71, 70, 69, 68, 67] (+0.0516).
- Judgment found a stable regime at PRE and between blocks and chose no intervention both times, so arm E ran with the same layout as C and no DD.
- E − C therefore compares two identical configurations, yet it reads −0.0059 (z = −2.2, 0/4). This is an accidental null control: it shows variation of about 0.006 AR between identical arms beyond shot noise, so a shot-noise z overstates certainty for differences of that size. Effects at or below about 0.006 should not be interpreted.
- DD forced on slightly lowered AR (−0.0066), which fits a case where the chosen qubits were already healthy (PRE snapshot s_rel of arm C: 0.928).
- The live-reaction gain seen on ibm_marrakesh did not appear here because no intervention was triggered. That is the designed behavior in a stable regime, not evidence for or against the reaction.

Job IDs: PRE db69kg4vf2bc73d0si1g; block 1 db69klamb58s738bfei0; block 2 db69koclf4us73c50tc0; POST db69ksamb58s738bferg.

| Ablation device | AR naive (A) | AR full EQG (E) | E − A | C − A | D − C | E − C |
|---|---|---|---|---|---|---|
| ibm_marrakesh | 0.7189 | 0.8126 | +0.0937 | +0.0736 | −0.0027 | +0.0200 |
| ibm_kingston | 0.7700 | 0.8157 | +0.0458 | +0.0516 | −0.0066 | −0.0059 |

Across both ablations the consistent, large contributor is layout selection by calibration. The DD and live-reaction contributions depend on the device and the day, which is why the next phase tests them with replicas rather than assuming them.

## 3. Limits of the Current Evidence

- Three backends (ibm_fez, ibm_marrakesh, ibm_kingston) and one workload (QAOA MaxCut, N = 8, p = 2).
- Four runs in total (two on ibm_fez, one ablation each on ibm_marrakesh and ibm_kingston), each in a single calibration cycle; no replica series across days yet.
- Sequential arms: snapshots mitigate chip drift but do not eliminate it.
- State-switching observed on a single qubit on a single day.
- No formal ablation (MORPH-only, DD-only, full stack) as separate arms; attribution comes from snapshots.
- Thresholds and the DD sequence are not shown to be optimal.
- No evidence yet on queue latency, multi-tenant use, or continuous operation.

These results establish viability of the mechanism, not a general product certification.

## 4. Reproducibility Protocol for Other Hardware

The same protocol applies on every platform so results are comparable.

1. **Pre-registration.** Fix the workload, shot plan, budget per arm, and the significance threshold ($z \geq 2$ on $\Delta$AR) before running.
2. **Crossed arm order.** Run both EQG-first and baseline-first, with PRE/MID/POST simultaneous snapshots, as in R1 and R2.
3. **Replica series.** Repeat each condition across multiple days and calibration cycles [proposed: at least 5 replicas per backend, to be confirmed].
4. **Three-arm ablation.** Layout-only, DD-only, and full stack against baseline.
5. **Publish records.** Job IDs, raw counts, report.json, and analysis scripts for every run.
6. **Negative control.** Run the stack on a noisy simulator and with the intervention disabled, to verify that no effect appears when none is expected.

## 5. Validation Matrix by Hardware Type

This matrix is a projection, not a result. For each platform the role adaptation is a hypothesis to be tested, and the acceptance criterion is the same pre-registered one.

| Hardware type | Expected dominant noise | Role adaptation to test | Validation focus | Acceptance |
|---|---|---|---|---|
| Other IBM superconducting devices | TLS defects, dephasing, crosstalk | Minimal changes: reuse AOM, MORPH, DD | Replicate ibm_fez effect on additional devices and calibration cycles | $z \geq 2$, same sign in both arm orders |
| Other superconducting vendors | Similar, with different gate sets and timing control | Re-implement DD insertion and health sensing in the vendor's SDK; remap native gates | Whether DD insertion is possible at the vendor's timing granularity; layout gain on their topology | $z \geq 2$ and ablation shows each active role contributes |
| Trapped ions | Magnetic-field drift, motional heating; long coherence | Layout role largely moot with all-to-all connectivity; shift weight to health sensing and echo-type DD | Whether Judgment avoids unnecessary intervention; gain from DD or recalibration triggers | $z \geq 2$, or documented "no benefit" with the hold decision working correctly |
| Neutral atoms | Laser and positional noise, atom loss | Replace layout with atom-placement or zone selection; spin-echo-style DD | Health signal definition; loss-aware execution | $z \geq 2$ on a task-appropriate figure of merit |
| Photonic and other | Loss, source quality, distinguishability | Redefine health metrics; DD likely not applicable | Define applicability before any claim | Applicability study first; claims only if a measurable effect is found |

A negative or null result on a platform is itself a valid outcome and will be reported with the same transparency.

## 6. Phased Reproducibility Plan

| Phase | Scope | Output |
|---|---|---|
| R0 | Replica series and formal ablations on ibm_fez | Replica statistics, effect-size confidence intervals, ablation attribution |
| R1 | Additional IBM devices (first data points: ibm_marrakesh, Section 2.7; ibm_kingston, Section 2.8) | Cross-device consistency of ΔAR and layout/DD contributions |
| R2 | At least one non-IBM superconducting platform | SDK adaptation notes and a first cross-vendor result |
| R3 | One non-superconducting platform | Applicability map: where each role helps, is unnecessary, or fails |
| R4 | Extra workloads (beyond QAOA MaxCut), larger N, and operational metrics | Overhead, queue-latency, and continuous-operation data |

Each phase ends with a report in the same format as this one, listing results, limits, and the go/no-go decision for the next phase. Platform access, timelines, and budgets for R2 and R3 depend on vendor availability and are [to be defined].

## 7. Conclusion

On ibm_fez, EQG produced a statistically significant AR improvement in two runs with inverted arm order, and snapshots attribute the gain to both layout selection and DD. The next step is not a wider claim but a reproducibility program: replicas and ablations first, then other devices and vendors, with the same pre-registered criteria and an honest record of where the mechanism does and does not apply.

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